

F1 CardioSentinel as an intelligent physical system

Four layers. Each band names what runs; the line beneath it names the evidence that constrains it.

Layer 4 AGENTIC

Evidence Agent · Explanation Agent · Research Assistant

claim boundary on every output — 18 of 25 Appendix A claims machine-checked



Layer 3 EVIDENCE

AlertEvent → EvidenceRecord → EvidenceGraph

35 nodes / 39 edges per alert · closed node kinds and edge relations



Layer 2 EDGE RUNTIME

StreamingInferenceSession, five pieces of causal state

~61× real time on a laptop CPU · encoder 4.161 ms/window median



evidence flows up; nothing flows back down

Layer 1 SIGNAL

StreamingPreprocessor → CausalWindowGenerator

→ 146-d representation → M1L / M2-G → U1 → T2 → T1

Train / runtime equivalence audit — the bridge, and the only real unknown

physiology half (18-d) bit-exact, 0.000e+00 on 64 of 64 audited rows · embedding half (128-d) max 7.15e-07 = 6 ULP of float32